

ECO387C - Fall 2024

Problem Set 5

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

This problem set asks you to implement value function iteration on your computer from scratch.

Dynamic Programming - Hiring and Firing in France

Consider a firm that produces using only labor, so that $Q = F(N) = N^{1-\alpha}$, where N is the number of current employees and $F' > 0$ and $F'' < 0$. In each period, a constant fraction of the workforce leaves due to exogenous separation. Let δ be the fraction of the workforce that leaves each period. The wage w is constant, but the price (p) of the firm's output is stochastic, following a three-state Markov chain: $p \in \{p_1, p_2, p_3\}$

$$\mathbf{P} = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix} = \begin{bmatrix} q & \frac{1-q}{2} & \frac{1-q}{2} \\ \frac{1-q}{2} & q & \frac{1-q}{2} \\ \frac{1-q}{2} & \frac{1-q}{2} & q \end{bmatrix}$$

For example, $P_{12} = \frac{1-q}{2}$ refers to the probability of transitioning from p_1 in the current period to p_2 in the next period. The firm cannot lay off its workers. It can only choose $N' \geq (1-\delta)N$. To formulate the firm's problem more formally, the firm seeks to maximize the present discounted value of profits:

$$\max_{\{h_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t [p_t Q_t - w N_t]$$

subject to

- Production function: $Q_t = N_t^{1-\alpha}$
- Evolution of workforce: $N_t = (1-\delta)N_{t-1} + h_t$
- Non-negativity constraint: $h_t \geq 0, \forall t$
- Initial condition: $N_{-1} > 0$

Answer the following questions:

1. What are the relevant state and control variables?

2. Write down the Bellman equation.
3. Solve the firm's problem using value function iteration and plot both the value function and the policy functions (N_{t+1} and h_t). Use the parameter values in the following table:

δ	$1+r$	w	q	α	$N_{\min}$	$N_{\max}$	p_1	p_2	p_3
0.05	1.03	0.4	0.9	0.3	0.01	1	0.4	0.45	0.5

$N_{\min}$ and $N_{\max}$ specifies the lower and upper bound of the interval for N .

4. Compute the value and policy functions without the "no-firing" constraint, where N_t can be chosen freely (i.e. h_t can be negative). What effect does the "no-firing" constraint have?